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AI for Bharat Hackathon

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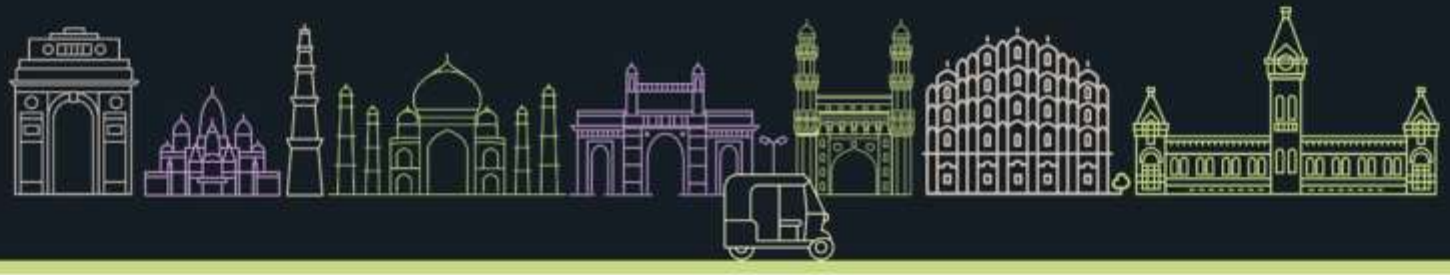


Team Name : Synthetix

Team Leader Name : Sarthak Deodhar

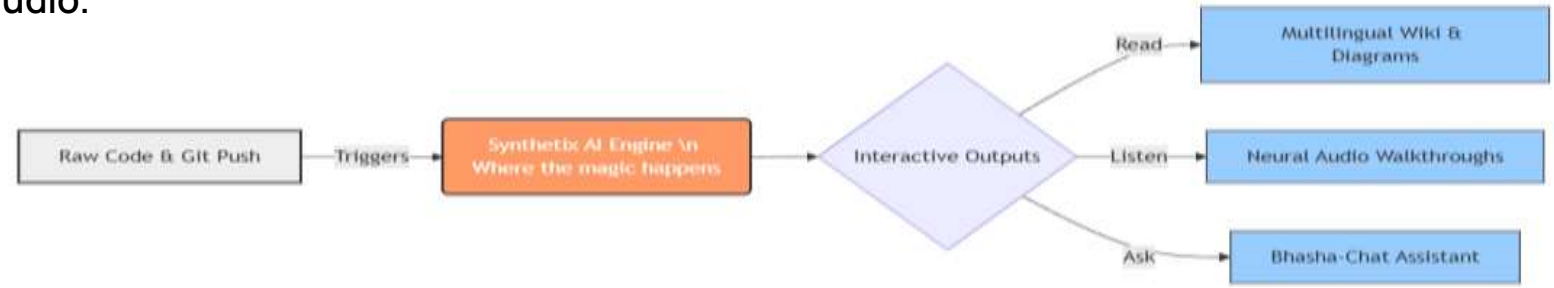
Problem Statement : Documentation Debt: Technical debt consumes 30% of senior developer capacity, costing teams significantly in lost productivity.
Linguistic Exclusion: English-only documentation creates barriers for Bharat's 5.5 million+ developers, particularly in Tier-2/3 cities.

Synthetix Docs: AI-Augmented Technical Navigator for Bharat

- **Concept:** An intelligent, serverless platform designed to transform static technical documentation into an interactive, multilingual learning assistant for the Indian developer ecosystem.

- **Core Innovation:** Unlike traditional wikis, the system is **event-driven**—it automatically regenerates documentation, architecture diagrams, and audio walkthroughs instantly whenever code is pushed to GitHub.

- **Impact:** It bridges the linguistic gap by enabling developers to query technical concepts in **20+ Indic languages** and receive context-aware, localized explanations via text or neural audio.



- ## How different is it from any of the other existing ideas?

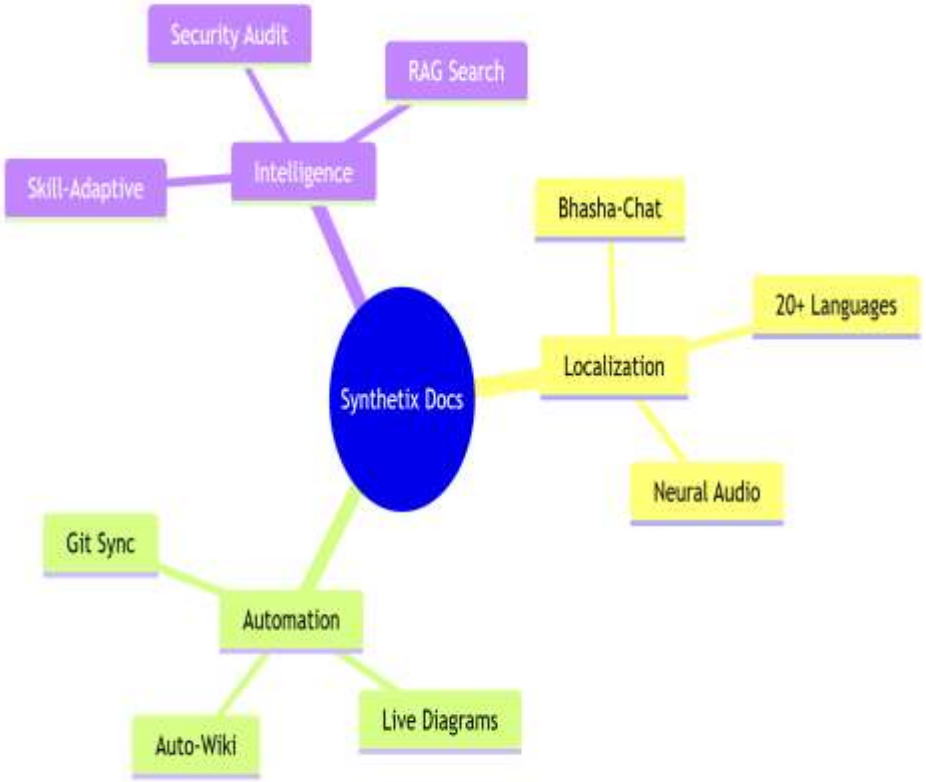
  - Unlike traditional static documentation that becomes obsolete quickly, our system is **event-driven and self-updating**. While standard wikis rely on manual edits, Synthetix Docs automatically regenerates technical guides, architecture diagrams, and security advisories immediately upon every Git push, ensuring 100% freshness.

- ## How will it be able to solve the problem?

  - It solves **"Linguistic Exclusion"** by enabling developers to query codebases in **20+ Indian languages** via the Bhasha-Chat RAG widget, breaking the English barrier.
  - It solves **"Onboarding Inefficiency"** by using **Skill-Adaptive Content** that personalizes documentation depth, reducing new hire ramp-up time from 3–4 weeks to just 3–5 days (80% reduction).

- ## USP of the proposed solution

  - **The Bharat Explainer**: The first tool to generate 60-second **neural audio walkthroughs** of complex code logic in 10+ Indic languages (Hindi, Marathi, Tamil, etc.) for deeper comprehension.
  - **Suraksha-Audit**: An integrated scanner that doesn't just document code but proactively identifies vulnerabilities and writes **Security Advisories** automatically.



Multilingual Semantic Search: Query technical documentation in **20+ Indian languages** with >90% accuracy using Amazon Bedrock.

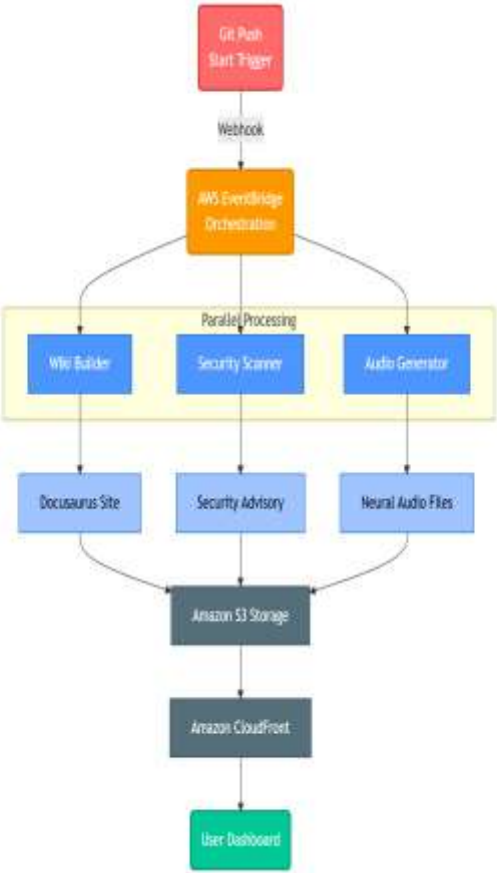
Autonomous Wiki & Mapper: Automatically regenerates the entire documentation portal and live **Mermaid.js architecture diagrams** within 5 minutes of a Git push.

The Bharat Explainer: Generates 60-second **neural audio walkthroughs** of code logic in 10+ Indic languages (Hindi, Marathi, Tamil, etc.).

Bhasha-Chat Widget: An embeddable **RAG-based chat widget** that allows developers to ask "What does this code do?" in their native language.

Suraksha-Audit: A security scanner that detects vulnerabilities (OWASP Top 10, secrets) and automatically writes **Security Advisories**.

Skill-Adaptive Content: A dynamic toggle that switches documentation depth between **"Junior"** (step-by-step) and **"Senior"** (architecture patterns) views.



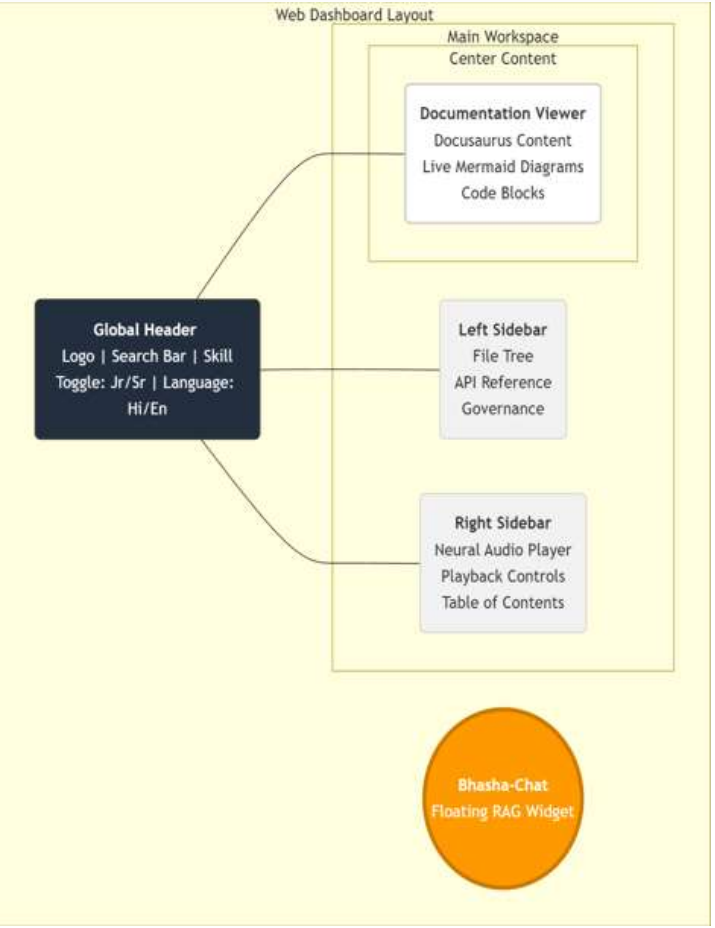
- **Trigger:** The process initiates instantly via a **GitHub Webhook** whenever code is pushed.
- **Orchestration: AWS EventBridge** receives the event and triggers three parallel Lambda functions.
- **Parallel Processing:**

1. **Wiki Builder:** Generates documentation and live architecture diagrams.
2. **Security Scanner:** Runs CodeGuru scans and identifies vulnerabilities.
3. **Audio Generator:** Synthesizes neural audio walkthroughs using Amazon Polly.

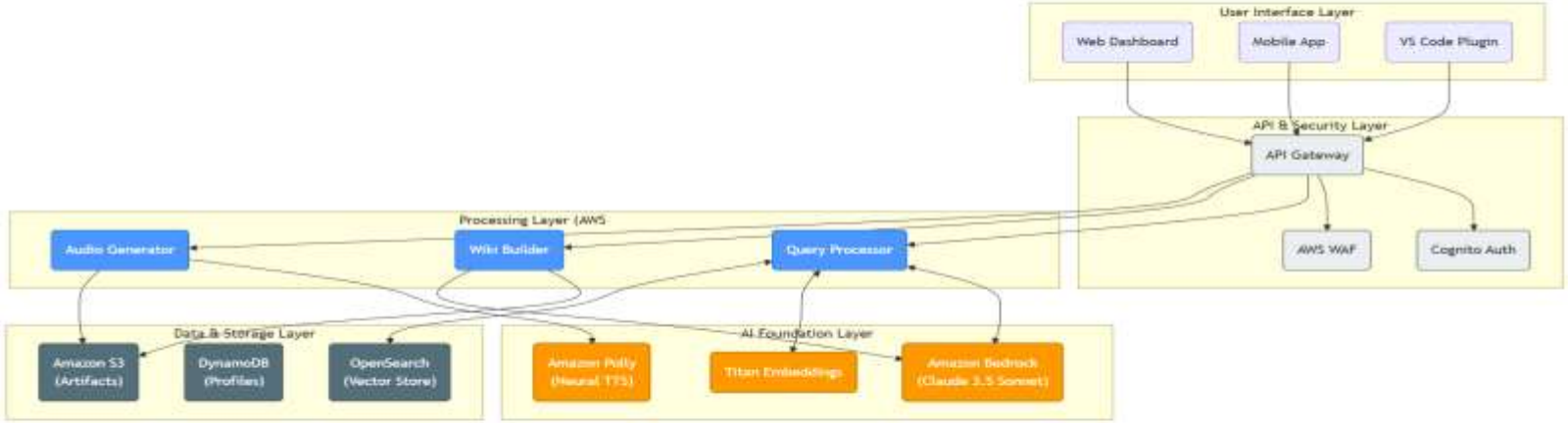
- **Delivery:** All artifacts are stored in **Amazon S3** and delivered globally via **CloudFront** to the user dashboard.

Wireframes/Mock diagrams of the proposed solution:

- **Unified Dashboard**: A single-page React application containing the Wiki, Interactive Diagrams, and Audio Player in one view.
- **Smart Header**: Contains the "**Skill Level Toggle**" (Junior/Senior) and **Language Selector** for instant personalization.
- **Bhasha-Chat Widget**: A collapsible, floating widget in the bottom-right corner for real-time, localized Q&A without leaving the documentation.
- **Audio Sidebar**: Dedicated controls for playback speed (0.75x - 1.5x) and language selection for the neural audio walkthroughs.



Architecture diagram of the proposed solution:



Serverless & Event-Driven: Built entirely on **AWS Lambda** and **EventBridge**, ensuring zero idle costs and infinite scalability for burst traffic.

Multi-Agent AI Layer: Leverages **Amazon Bedrock (Claude 3.5 Sonnet)** for reasoning, **Amazon Polly** for neural audio synthesis, and **Titan Embeddings** for vector search.

High-Performance Data Layer: Uses **OpenSearch Serverless** for millisecond-latency vector retrieval (RAG) and **DynamoDB** for storing user profiles and learning paths.

Technologies to be used in the solution:

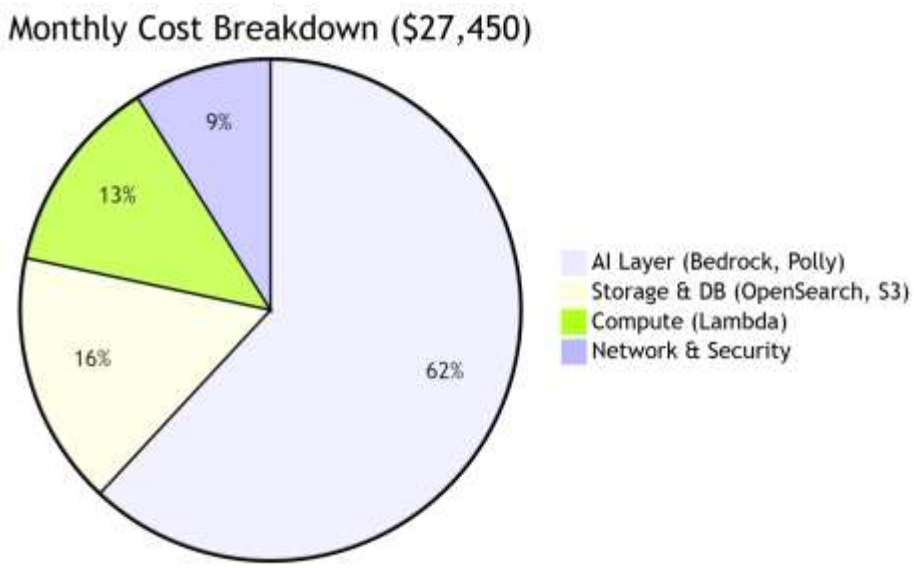
- **Generative AI & ML:**
- **Amazon Bedrock:** Claude 3.5 Sonnet (Reasoning) & Titan Embeddings (Vector Search).
- **Amazon Polly:** Neural Text-to-Speech for localized audio.
- **Amazon Translate:** Real-time multilingual translation.

- **Serverless Compute & Integration:**
- **AWS Lambda:** Serverless function execution.
- **Amazon EventBridge:** Event-driven orchestration.

- **Data & Storage:**
- **Amazon OpenSearch Serverless:** High-performance vector database for RAG.
- **Amazon DynamoDB:** User profiles and learning paths.
- **Amazon S3:** Object storage for documentation artifacts and audio files.

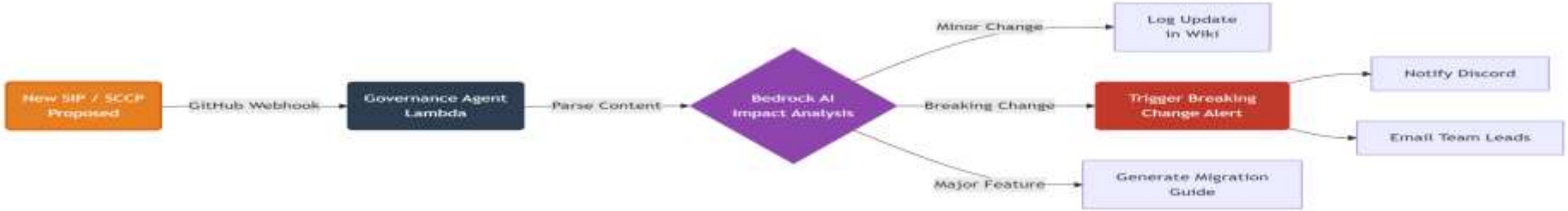
- **Security & Frontend:**
- **Amazon CodeGuru Security:** Automated vulnerability scanning.
- **React.js & Docusaurus:** Frontend UI and static site generation.

Estimated implementation cost (optional):



- **AI Services (~62%): $17,000** – Primarily Amazon Bedrock (Claude 3.5 Sonnet) and Polly for high-quality reasoning and neural audio.
- **Data & Storage (~16%): $4,500** – OpenSearch Serverless for vector RAG and DynamoDB for user profiles.
- **Compute (~13%): $3,500** – AWS Lambda functions for event-driven processing.
- **Optimization Strategy:** Projected to reduce costs by **32%** (to ~$18,750/month) within 6 months by implementing aggressive caching and AWS Savings Plans.

- Carbon-Aware Architecture: Integrates with the AWS Customer Carbon Footprint Tool to track real-time emissions and optimize regional workload distribution (e.g., routing traffic to regions with higher renewable energy availability).
- Eco-Friendly Compute: Utilizes ARM-based AWS Graviton processors (20% more efficient) and schedules heavy batch jobs (like audio generation) during off-peak, low-carbon intensity hours.
- Sustainability Goals: Targeting < 1.5 kg CO2e per user/month and achieving 40% renewable energy usage in the first year of operation.



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